

Ex.no:4a	EXPLORATORY DATA ANALYSIS (EDA) – DATA INSPECTION, FILTERING, STATISTICAL ANALYSIS AND CORRELATION

Aim:

To perform data inspection, filtering, grouping, aggregation, descriptive statistical analysis, and correlation analysis using Pandas and visualize the results using Matplotlib.

Algorithm:

1. Import the required Pandas, NumPy and Matplotlib libraries.
2. Read the stud_data.csv file using Pandas.
3. Display the dataset and inspect the first few records.
4. Find the number of rows and columns and display the data types.
5. Generate the statistical summary of the dataset.
6. Filter the students based on marks and attendance conditions.
7. Sort the student records based on marks and attendance.
8. Group the data based on department and calculate average marks and attendance.
9. Calculate mean, median, minimum, maximum and standard deviation of marks.
10. Calculate the correlation between marks and attendance.
11. Create bar chart, line chart, scatter plot, histogram and box plot using Matplotlib.
12. Display the results and interpret the obtained data.

Program:

```
# Import libraries
```

```
import pandas as pd
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
# Read dataset
```

```
df = pd.read_csv("stud_data.csv")
```

```
# Display dataset
```

```
print(df)
```

```
# Display first five records
```

```
print(df.head())
```

```
# Display shape
```

```
print(df.shape)
```

```
# Display data types
```

```
print(df.dtypes)
```

```
# Display statistical summary
```

```
print(df.describe())
```

```
# Filter students with marks greater than 80
```

```
print(df[df["Marks"] > 80])
```

```
# Filter students with attendance greater than 90
```

```
print(df[df["Attendance"] > 90])
```

```
# Sort students by marks
```

```
print(df.sort_values("Marks", ascending=False))
```

```
# Sort students by attendance
```

```
print(df.sort_values("Attendance", ascending=False))
```

```
# Department-wise average marks
```

```
print(df.groupby("Department")["Marks"].mean())
```

```
# Department-wise average attendance
```

```
print(df.groupby("Department")["Attendance"].mean())
```

```
# Calculate mean
```

```
print("Mean Marks:", df["Marks"].mean())
```

```
# Calculate median
```

```
print("Median Marks:", df["Marks"].median())
```

```
# Calculate minimum
```

```
print("Minimum Marks:", df["Marks"].min())
```

```
# Calculate maximum
```

```
print("Maximum Marks:", df["Marks"].max())
```

```
# Calculate standard deviation
```

```
print("Standard Deviation:", df["Marks"].std())
```

```
# Display correlation matrix
```

```
print(df[["Marks", "Attendance"]].corr())
```

```
# Display correlation between marks and attendance
```

```
print("Correlation:", df["Marks"].corr(df["Attendance"]))
```

```
# Bar chart
```

```
plt.bar(df["Name"], df["Marks"])
```

```
plt.title("Student Marks")
```

```
plt.xlabel("Name")
```

```
plt.ylabel("Marks")
```

```
plt.show()
```

```
# Line chart
```

```
plt.plot(df["Name"], df["Attendance"], marker="o")
```

```
plt.title("Student Attendance")
```

```
plt.xlabel("Name")
```

```
plt.ylabel("Attendance")
```

```
plt.show()
```

```
# Scatter plot
```

```
plt.scatter(df["Attendance"], df["Marks"])
```

```
plt.title("Marks vs Attendance")
```

```
plt.xlabel("Attendance")
```

```
plt.ylabel("Marks")
```

```
plt.show()
```

```
# Histogram
```

```
plt.hist(df["Marks"], bins=5)
```

```
plt.title("Marks Distribution")
```

```
plt.xlabel("Marks")
```

```
plt.ylabel("Number of Students")
```

```
plt.show()
```

```
# Box plot
```

```
plt.boxplot(df["Marks"])
```

```
plt.title("Marks Box Plot")
```

```
plt.ylabel("Marks")
```

```
plt.show()
```

Output:

```
Roll No    Name Department Marks Attendance
0    101    Arun         CSE      85         92
1    102    Bala         IT       78         88
2    103    Divya        CSE      95         96
3    104    Priya        AI&DS    90         94
4    105    Ravi         IT       72         85
5    106    Karthik      AI&DS    88         91
```

```
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4    105    Ravi         IT       72         85
```

(6, 5)

```
Roll No      int64
Name         object
Department   object
Marks        int64
Attendance   int64
dtype: object
```

```
count      Roll No      Marks Attendance
mean    103.500000    84.666667      91.00
std       1.870829      8.382521       4.00
min      101.000000    72.000000      85.00
25%      102.250000    79.750000      88.75
50%      103.500000    86.500000      91.50
75%      104.750000    89.500000      93.50
max      106.000000    95.000000      96.00

Roll No    Name Department Marks Attendance
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4	105	Ravi	IT	72	85

Department

AI&DS 89.0

CSE 90.0

IT 75.0

Name: Marks, dtype: float64

Department

AI&DS 92.5

CSE 94.0

IT 86.5

Name: Attendance, dtype: float64

Mean Marks: 84.66666666666667

Median Marks: 86.5

Minimum Marks: 72

Maximum Marks: 95

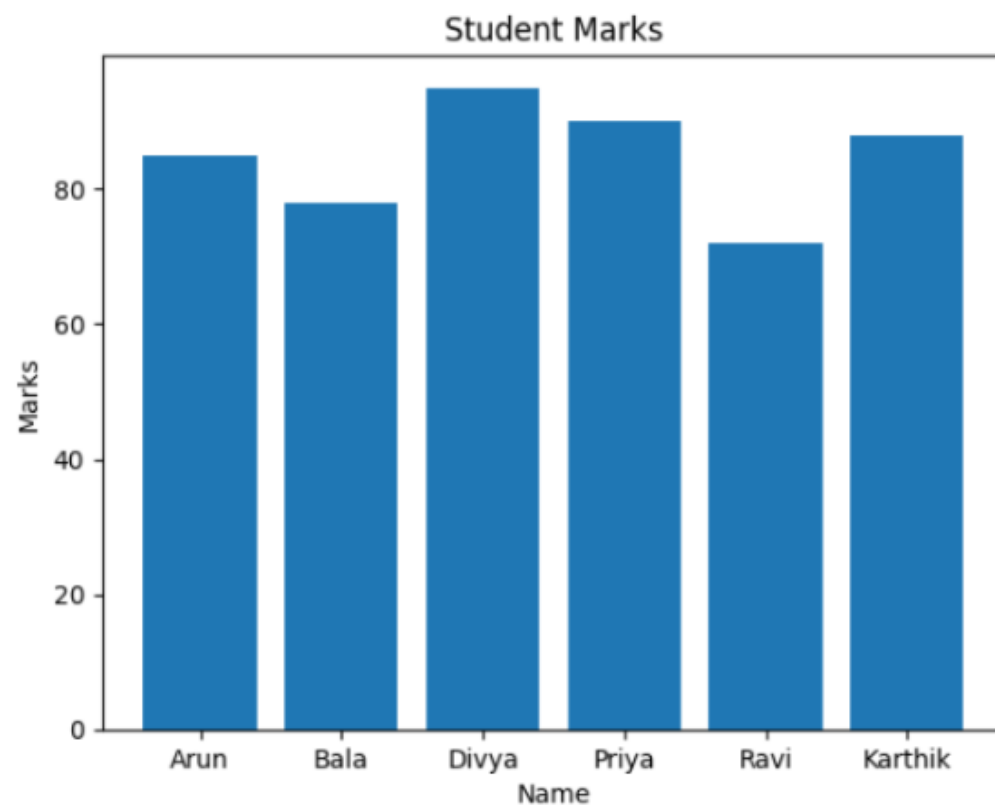
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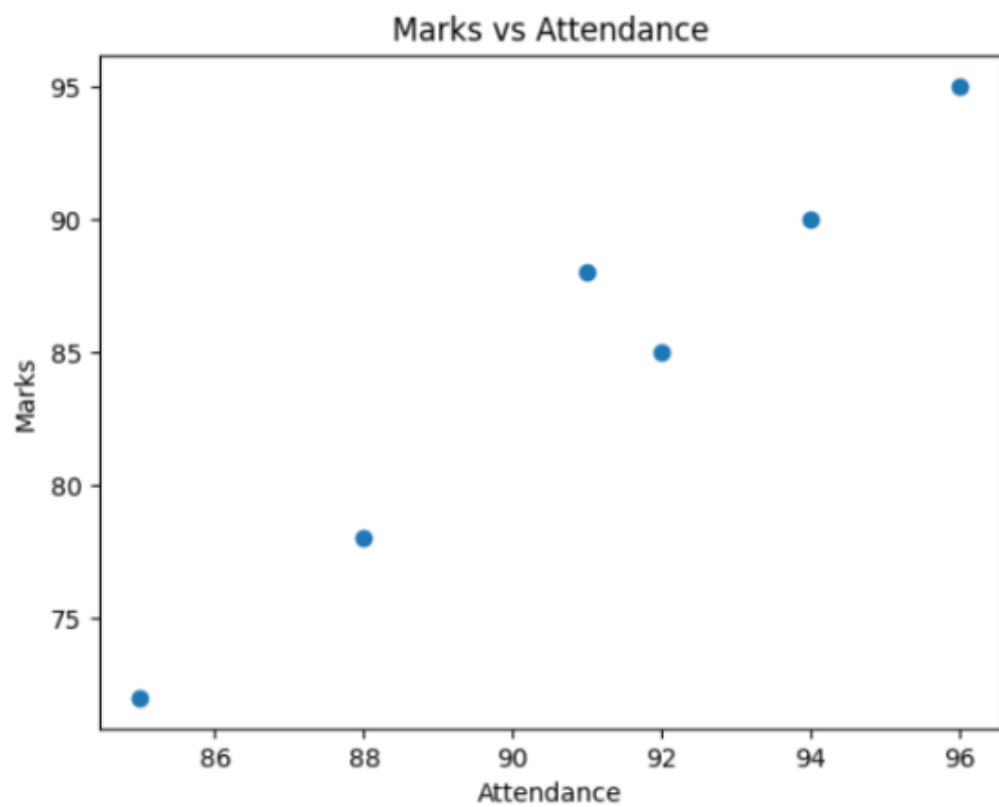
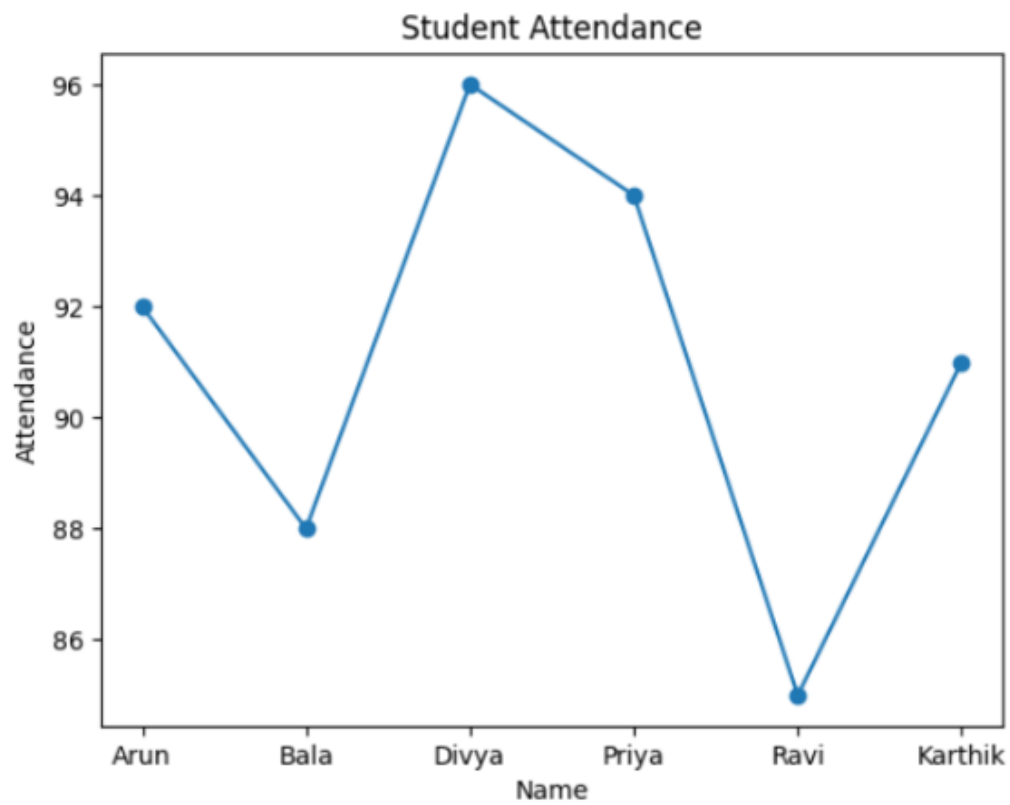
	Marks	Attendance
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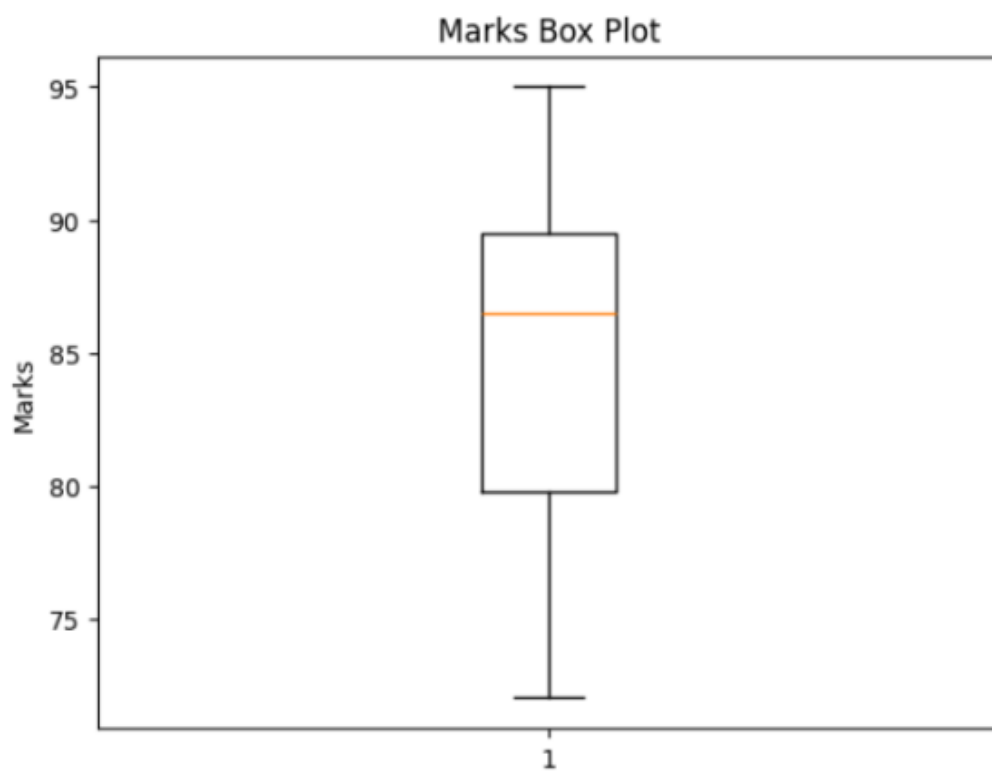
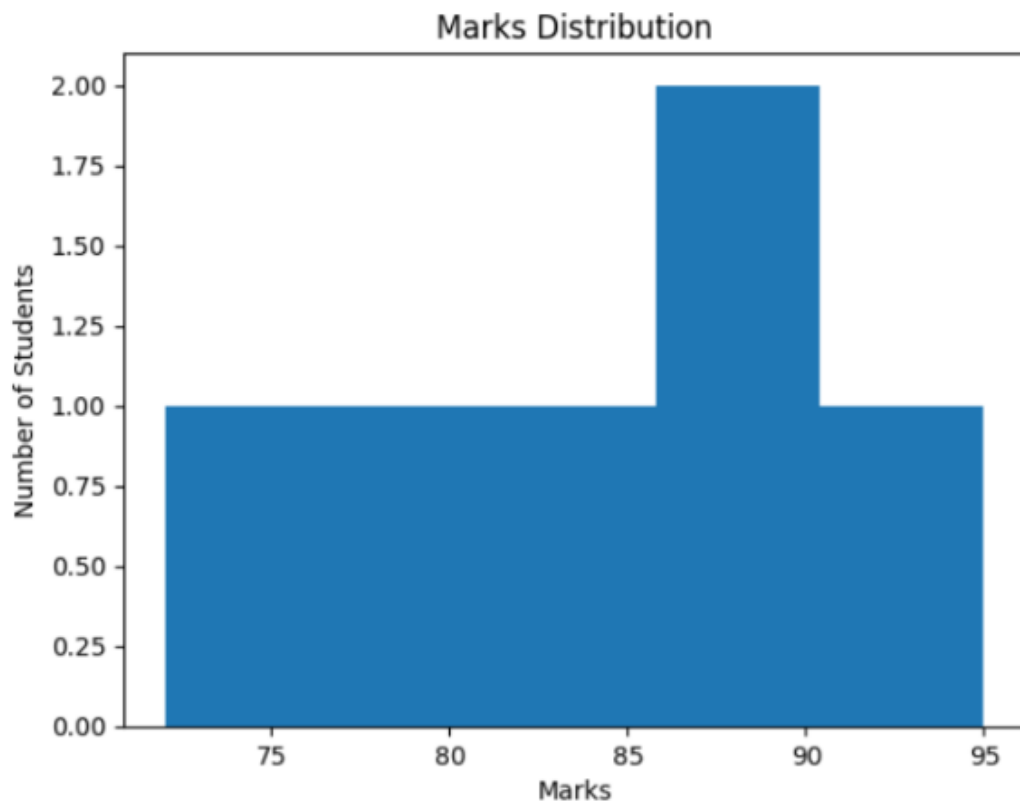
Marks	1.000000	0.978226
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Attendance	0.978226	1.000000
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Correlation: 0.9782259433305447







Result: